

EU Digital Product Passport update

In April 2025, the European Commission published the *Ecodesign for Sustainable Products Regulation (ESPR) Work Programme*¹ for 2025–2030 and it opened a consultation on the rules for DPP service providers².

This paper intends to provide a brief update on the GS1 input to the legislative and standardisation activities.

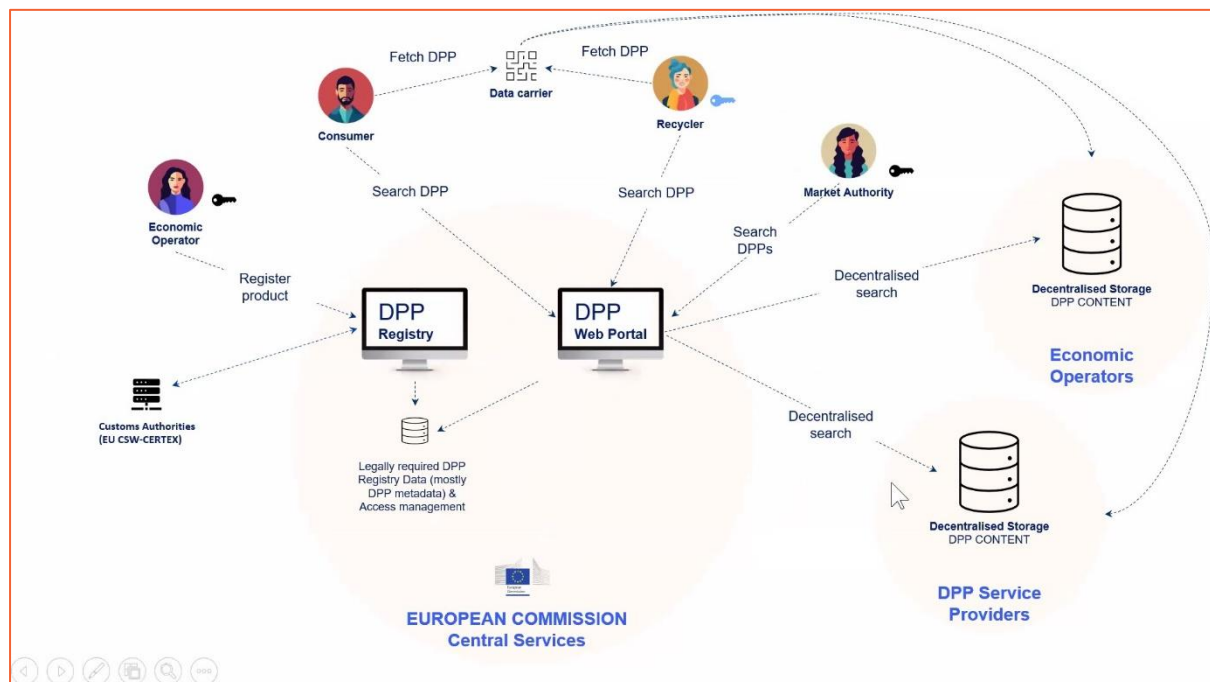


Figure 1 – DPP system building blocks (TBC)

Among several ambitious goals and new requirements, the EU Ecodesign for Sustainable Products Regulation³ (ESPR) introduces the Digital Product Passport (DPP) for products, components and intermediates products being placed on the EU market (manufactured in Europe or exported into the EU), impacting companies from all over the world, customs and cross border operations.

¹ Ecodesign for Sustainable Products and Energy Labelling Working Plan 2025-2030 available here: <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A52025DC0187&qid=1744814743855>

² Digital Product Passport – rules for service providers: https://ec.europa.eu/info/law/better-regulation/have-your-say/initiatives/14382-Digital-product-passport-rules-for-service-providers_en

³ Regulation (EU) 2024/1781 of the European Parliament and of the Council of 13 June 2024 establishing a framework for the setting of ecodesign requirements for sustainable products: <https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A32024R1781&qid=1719580391746>

The ESPR entered into force on 18 July 2024 as a key pillar of the EU Circular Economy agenda towards the EU climate neutrality goals and several EU delegated and implementing acts are going to be issued to concretely introduce the DPP into sectors.

The ESPR work programme confirmed the first wave of product groups that will be targeted for the ecodesign requirements and DPP implementation which are *textiles and apparel*, *mattresses*, *furniture*, *tyres*, *iron and steel*, *aluminium*, and 16 *energy related products*. The selection reflects a focus on sectors with high environmental impact and strong potential for circularity and sustainability improvements as highlighted in the EU Joint Research Centre study⁴.

The European Commission shared recently an overview of the DPP building blocks and responsibilities that will be necessary for completing the DPP ecosystem⁵ and ensure its functioning.

In accordance with the requirements set out in ESPR Article 19⁶, the European Commission established the Ecodesign Forum where pre-selected stakeholders are being consulted, including GS1, on additional upcoming rules, ecodesign requirements and working plans.

Implementing and Delegated acts



Figure 2 – EU timeline for delegated acts and implementing acts

The delegated acts will set ecodesign and energy labelling requirements on a product-by-product basis or for groups of similar products. The process will be based on thorough preparatory studies and impact assessments, with close involvement of stakeholders, including through the Ecodesign Forum.

⁴ <https://publications.jrc.ec.europa.eu/repository/handle/JRC138903>

⁵ See Figure 1 - to be confirmed, may vary.

⁶ https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX%3A32024R1781&qid=1719580391746#art_19



The European Commission has started working on the Delegated Act on DPP Service Providers, that will detail the rules and details needed for Service Providers to follow to store the DPP on behalf of the economic operators (ESPR Article 11).

Further to the inter-DG taskforce created within the European Commission on DPP, there are other different agencies and services at national level in EU Member States that are investigating the implications of this newly introduced structure. One example of many is the DATAPIPE project⁷, that was concluded at the beginning of 2025. The objective of the DATAPIPE project was to develop the 'extended data pipeline for the monitoring of the circular economy', which next to data about the goods and logistics data about the movement of goods, also covers Digital Product Passport data, such as data about the material composition of products, production processes, use, and recycling. Such data can be used for compliance monitoring of the circular economy.

GS1 Standards enabling the EU digital product passport

GS1 standards for identification, automatic identification and data capture (AIDC), and data sharing can support ESPR requirements of the European Union and other jurisdictions as well. To increase understanding of GS1 standards conformant systems, the GS1 White Paper⁸ explains the different aspects of how GS1 standards, widely implemented today or planned for implementation by 2027, can support the evolving circular economy with little to no disruption to industry.

The White Paper focuses on a web-enabled, structured path product identification (UPID⁹) syntax and can be summarised as follows:

- consumers use the smartphone to scan an AIDC carrier, land on a Web page, and then click on different sources of information per their interest (e.g., DPP data, instructions for use).
- by using these identification elements independent of the domain address and never reallocating them once assigned, the persistence of product identification is assured
- all of this occurs by encoding the domain address of the manufacturer before the GS1 identifiers in a structured, web-enabled syntax.

The paper also focuses also on the discussion of Economic Operator ID (EOID), Facility ID (FID) and GS1 Data Sharing standards.

GS1's collaboration with industry associations on DPP and Digital Labelling

GS1 has established a close collaboration with EU trade and sector associations part of the ESPR prioritised sectors¹⁰ (e.g. Cosmetics Europe, AIM, EuroCommerce, AISE). In this collaboration, GS1 contributes its technical expertise, complementing the advocacy and industry-specific insights provided by the associations.

⁷ DATAPIPE project: <https://www.tudelft.nl/tbm/onderzoek/projecten/datapipe-project>

⁸ GS1 Standards enabling the EU digital product passport: <https://gs1.eu/wp-content/uploads/2024/12/GS1-Standards-Enabling-DPP.pdf>

⁹ Unique Product ID

¹⁰ ESPR Article 18



Complementary to the work done with each association, GS1 participates as well in the *call for a digital transition for consumer information*¹¹ put forward by the Digital Consumer Information Alliance (DCIA). Together with EuroCommerce and Colruyt, on the 8th April 2025 GS1 joined the exhibition event organised by the Alliance to present to EU policymakers the latest updates on digital labelling conveying one message: whilst essential product details should always remain on-pack, integrating digital labelling aligns with the digital era and unlocks numerous benefits for both businesses and consumers.

GS1's involvement in major policy and standardisation efforts (internal and external)

GS1 Circularity/DPP Mission Specific Work Group

The GS1 GSMP (Global Standards Management Process) is a community-based forum for businesses facing similar problems to work together and develop standards-based solutions. Following the ESPR proposal, GS1 opened already in April 2023 a global, open-to-all standardisation table set up to support industry's identification and data sharing needs related to Circularity/DPP. To join the Mission Specific Working Group on Circularity/DPP, you can follow [this link](#) and be redirected to the registration page.

CEN/CENELEC JTC-24

GS1 in Europe has established a liaison with the technical committee JTC-24 that was given mandate by the European Commission to develop the standards for DPP, and is currently following the process and giving its contribution.

ISO/PWI 25534-1 "Digital Product Passport—Part 1: Overview and Fundamental Principles"

The ISO TC/154 JWG9 is a joint working group overseeing all the projects that require a collaboration between ISO and UNECE¹². In this context, the ISO/PWI 25534-1 is a project launched on the 25th February 2025 aiming to create a global approach to DPP.

The declared scope of the project, intended to be Part 1 of a DPP standards series, is to provide a comprehensive overview and establish core principles for developing a global DPP standard system. It aims to foster a unified understanding of DPP architecture at the international level and support subsequent standardization efforts.

GS1 was already part of JWG9, and is now actively participating in this new ISO/UNECE working group.

CIRPASS

Funded by the European Commission under the Digital Europe Programme, CIRPASS¹³ was a collaborative initiative to prepare the ground for the gradual piloting and deployment of a standards-based Digital Product Passport (DPP) aligned with the

¹¹ DCIA communication: <https://www.eurocommerce.eu/app/uploads/2024/09/2024-09-23-a-call-for-a-digital-transition-for-consumer-information-by-dcia-2024.pdf>

¹² The only exception is EDIFACT standard, ISO 9735, who has historically been managed by a dedicated group, JWG1.

¹³ <https://cirpassproject.eu/>



requirements of the Proposal for Ecodesign for Sustainable Product Regulations (ESPR), with an initial focus on the electronics, batteries, and textile sectors.

The project consortium was composed of 31 partners representing thousands of industrial, research, digital, and international, standards, organisations across Europe and beyond. The 18-month project responded to the European Commission's call on creating a clear concept for the DPP, defining a cross-sectoral product data model and DPP system with demonstrated benefits for the circular economy as well as developing roadmaps for its deployment, that ended in May 2024.

CIRPASS 2

Shortly after the end of CIRPASS, CIRPASS-2¹⁴ was launched. CIRPASS-2 is an Innovation Action project funded by the European Commission's Digital Europe Programme that will be ongoing from May 2024 until April 2027. The project counts on a large consortium of organisations from all over Europe and beyond. With 13 lighthouse pilots, the project will demonstrate functioning DPPs in real settings and at scale in **four target value chains: textiles, electrical and electronic equipment, tyres and construction materials.**

The project will also create a wide community of DPP stakeholders to facilitate the deployment of DPPs in different product sectors, across Europe and beyond.

In both projects, GS1 in Europe collaborated actively on developing the CIRPASS and CIRPASS-2 DPP architecture; in addition, for CIRPASS-2, GS1 in Europe is sub-task leader of T3.2A that focuses on developing upper level sectoral ontologies, in particular for the textile sector.

For any questions, please contact the GS1 Global Office Public Policy team

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¹⁴ <https://cirpass2.eu/>